

Essentials in No Arbitrage and Probability - Part One

Computational Methods in Finance, Code 21040

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Agenda

- Financial Assets
- Stochastic Processes: A Toolkit
- Information
- Expectations
- Measurability and independence
- Martingales

The Market (discrete time) I

Time: $t = 0, \Delta t, 2\Delta t, \dots, T$

Securities:

- d risky securities with prices $S_i = \{S_i(t)\}_{t=0}^T$, for $i = 1, \dots, d$.
Return over $[t, t + \Delta t[$

$$\frac{S_i(t + \Delta t) - S_i(t)}{S_i(t)} \text{ is not known at } t$$

- The *locally riskless* security with price $B = \{B(t)\}_{t=0}^T$, whose price at time t is known at $t - \Delta t$. We also assume $B(0) = 1$.

The return

$$\frac{B(t + \Delta t) - B(t)}{B(t)} \text{ is known at } t$$

The Market (discrete time) II

Moreover, we denote with $r(t)$ the locally riskless interest rate for period $[t, t + \Delta t[$

$$r(t) \Delta t = \frac{B(t + \Delta t) - B(t)}{B(t)}$$

for $t = 0, \Delta t, \dots, T - \Delta t$.

The quantity $r(t)$ represents the net rate of return obtained by buying one unit of the security B at time t and reselling it at time $t + \Delta t$.
 $r(t)$ is *known* at t .

To describe the evolution of the (risky) assets:

- Ω set of events (states of nature): finite in discrete time.
 $\Omega = \{\omega_1, \dots, \omega_K\}$

The Market (discrete time) III

- $\mathbb{P}$ probability measure on Ω , $\mathbb{P}(\omega_k) > 0$ for all ω_k , $\mathbb{P}(\Omega) = 1$
(Investors know the probability distribution of the risky returns).
 $\mathbb{P}$ -*expected value* of $S_3(t)$:

$$\mathbb{E}^{\mathbb{P}}[S_3(t)] = \sum_{\omega_k \in \Omega} S_3(t)(\omega_k) \mathbb{P}(\omega_k)$$

- *Information structure*

Describing information evolution I

[DEF] We call *partition of Ω* a collection $\mathcal{A} = \{\mathcal{A}_1, \dots, \mathcal{A}_I\}$ of subsets $\mathcal{A}_j$ of Ω such that

$$\begin{aligned}\mathcal{A}_i \cap \mathcal{A}_j &= \emptyset \text{ for any } i \neq j \\ \bigcup_{i=1}^I \mathcal{A}_i &= \Omega\end{aligned}$$

[DEF] Comparison between partitions: finer and coarser partitions.

Given two partitions $\mathcal{A}$ and $\mathcal{A}'$, we say that $\mathcal{A}'$ is finer than $\mathcal{A}$ if every element of $\mathcal{A}'$ is contained in an element of $\mathcal{A}$. Formally, given $\mathcal{A}' = \{\mathcal{A}'_1, \dots, \mathcal{A}'_m\}$ and $\mathcal{A} = \{\mathcal{A}_1, \dots, \mathcal{A}_I\}$, we say that $\mathcal{A}'$ is finer than $\mathcal{A}$ if for all $\mathcal{A}'_j \in \mathcal{A}'$ there exists $\mathcal{A}_j \in \mathcal{A}$ such that $\mathcal{A}'_j \subseteq \mathcal{A}_j$.

Describing information evolution II

[DEF] Information Structure. Given the set of dates $\mathcal{T} = \{0, \Delta t, \dots, T - \Delta t\}$ and the set of states $\Omega = \{\omega_1, \dots, \omega_K\}$, we call information structure on $\mathcal{T}$ and Ω a family $\mathcal{P} = \{\mathcal{P}_t\}_{t=0}^T$ of partitions of Ω satisfying the following three properties:

- ① $\mathcal{P}_0 = \{\Omega\}$;
- ② $\mathcal{P}_{t+\Delta t}$ is finer than $\mathcal{P}_t$ for all $t = 0, \Delta t, \dots, T - \Delta t$;
- ③ $\mathcal{P}_T = \{\{\omega_1\}, \{\omega_2\}, \dots, \{\omega_K\}\}$

Notation: f_h^t element of $\mathcal{P}_t$

Describing information evolution III

[DEF] A random variable is known at t if X is $\mathcal{P}_t$ – measurable :

$$X(\omega_k) = X(\omega_j) \text{ for all } \omega_k, \omega_j \in f_h^t \text{ for all } f_h^t \in \mathcal{P}_t$$

Notation for $\mathcal{P}_t$ –measurable X :

- $X(f_h^t) \equiv X(\omega_k) = X(\omega_j)$ for all $\omega_k, \omega_j \in f_h^t$ for all $f_h^t \in \mathcal{P}_t$
- $X : \mathcal{P}_t \rightarrow \mathfrak{R}$

[DEF] A stochastic process

$$Y = \{Y(t)\}_{t=0}^T$$

where $Y(t) : \Omega \rightarrow \mathfrak{R}$, is adapted to $\mathcal{P}$ if $Y(t)$ is $\mathcal{P}_t$ – measurable for all t .

Updating the expectation I

$\mathbb{Q}$ (strictly positive) probability on Ω .

Random variable $X : \Omega \rightarrow \{x_1, \dots, x_M\} \subset \mathbb{R}$

$\mathbb{Q}$ —**(Unconditional) expectation** of X is

$$\mathbb{E}^{\mathbb{Q}}[X] = \sum_{\omega_k \in \Omega} X(\omega_k) \mathbb{Q}(\omega_k)$$

$\mathbb{Q}$ —**Conditional expectation** of X at t :

$$\mathbb{E}^{\mathbb{Q}}[X | \mathcal{P}_t](f_h^t) = ?$$

Conditional probability of ω_k given f_h^t is:

$$\mathbb{Q}[\omega_k | f_h^t] = \frac{\mathbb{Q}[\{\omega_k\} \cap f_h^t]}{\mathbb{Q}[f_h^t]} = \begin{cases} \frac{\mathbb{Q}[\omega_k]}{\mathbb{Q}[f_h^t]} & \text{if } \omega_k \in f_h^t \\ 0 & \text{if } \omega_k \notin f_h^t \end{cases}$$

Updating the expectation II

[DEF] $\mathbb{Q}$ -Conditional expectation of X at t is

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[X|\mathcal{P}_t](f_h^t) &= \sum_{\omega_k \in \Omega} X(\omega_k) \mathbb{Q}[\omega_k | f_h^t] \\ &= \sum_{\omega_k \in f_h^t} X(\omega_k) \frac{\mathbb{Q}[\omega_k]}{\mathbb{Q}[f_h^t]}\end{aligned}$$

Properties of the conditional expectation I

- $\mathcal{P}_0$ -conditional expected value simply reduces to the expected value of $X : \Omega \rightarrow \mathbb{R}$:

$$\mathbb{E}^{\mathbb{Q}}[X | \mathcal{P}_0](f_1^0) = \sum_{\omega_k \in \Omega} X(\omega_k) \mathbb{Q}[\omega_k | \Omega] = \mathbb{E}^{\mathbb{Q}}[X]$$

- **The conditional expectation of a measurable random variable is the random variable itself**

$X : \Omega \rightarrow \{x_1, \dots, x_M\} \subset \mathbb{R}$ is $\mathcal{P}_t$ -measurable :

$$X(\omega_k) = X(\omega_j) = X(f_h^t) \text{ for all } \omega_k, \omega_j \in f_h^t \text{ for all } f_h^t \in \mathcal{P}_t$$

Properties of the conditional expectation II

But then

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[X|\mathcal{F}_t](f_h^t) &= \sum_{\omega_k \in \Omega} X(\omega_k) \mathbb{Q}[\omega_k | f_h^t] \\&= \sum_{\omega_k \in f_h^t} \underbrace{X(\omega_k)}_{\text{constantly equal to } X(f_h^t)} \frac{\mathbb{Q}[\omega_k]}{\mathbb{Q}[f_h^t]} \\&= X(f_h^t) \sum_{\omega_k \in f_h^t} \frac{\mathbb{Q}[\omega_k]}{\mathbb{Q}[f_h^t]} \\&= X(f_h^t) \frac{1}{\mathbb{Q}[f_h^t]} \underbrace{\sum_{\omega_k \in f_h^t} \mathbb{Q}[\omega_k]}_{\mathbb{Q}[f_h^t]} = X(f_h^t)\end{aligned}$$

Properties of the conditional expectation III

- **The law of iterated expectations (Tower property).**

Given $0 \leq t_1 \leq t_2 \leq T$ the following holds:

$$\mathbb{E}^{\mathbb{Q}} \left[\mathbb{E}^{\mathbb{Q}} [X | \mathcal{P}_{t_2}] \middle| \mathcal{P}_{t_1} \right] = \mathbb{E}^{\mathbb{Q}} [X | \mathcal{P}_{t_1}].$$

- $\mathcal{P}_t$ —**measurable** random variables can be "taken out" of $\mathcal{P}_t$ -conditional expectations.

If $h(t)$ be measurable with respect to $\mathcal{P}_t$

$$\mathbb{E}^{\mathbb{Q}} [h(t) X | \mathcal{P}_t] = h(t) \mathbb{E}^{\mathbb{Q}} [X | \mathcal{P}_t]$$

- **Linearity of Conditional Expectation.**

If $X, Y : \Omega \rightarrow \mathfrak{R}$ and $h(t), k(t) : \mathcal{P}_t \rightarrow \mathfrak{R}$, then

$$\mathbb{E}^{\mathbb{Q}} [h(t) X + k(t) Y | \mathcal{P}_t] = h(t) \mathbb{E}^{\mathbb{Q}} [X | \mathcal{P}_t] + k(t) \mathbb{E}^{\mathbb{Q}} [Y | \mathcal{P}_t]$$

Independent random variables I

[DEF] A random variable $X : \Omega \rightarrow \{x_1, \dots, x_M\} \subset \mathfrak{R}$ is independent of the information $\mathcal{P}_t$ with respect to the probability $\mathbb{Q}$ if

$$\mathbb{Q} [\{\omega : X(\omega) = x_m\} \cap f_h^t] = \mathbb{Q} [\{\omega : X(\omega) = x_m\}] \cdot \mathbb{Q} [f_h^t]$$

for all $f_h^t \in \mathcal{P}_t$ and for all $m = 1, \dots, M$,
i.e. the subsets of Ω

$\{\omega \in \Omega : X(\omega) = x_m\}$ and $f_h^t \subset \Omega$ are independent w.r.t. $\mathbb{Q}$

for all $f_h^t \in \mathcal{P}_t$ and for all $m = 1, \dots, M$.

Independent random variables II

- If $X : \Omega \rightarrow \{x_1, \dots, x_M\} \subset \mathbb{R}$ is $\mathcal{P}_t$ **independent of the information $\mathcal{P}_t$ with respect to the probability $\mathbb{Q}$** it takes *all* possible realizations $x_1, \dots, x_M$ on *any* $f_h^t \in \mathcal{P}_t$:

$$\{\omega : X(\omega) = x_m\} \cap f_h^t \neq \emptyset \text{ for all } x_m = x_1, \dots, x_M$$

- If $X : \Omega \rightarrow \{x_1, \dots, x_M\} \subset \mathbb{R}$ is $\mathcal{P}_t$ **independent of the information $\mathcal{P}_t$ with respect to the probability $\mathbb{Q}$** , then

$$\mathbb{E}^{\mathbb{Q}} [X | \mathcal{P}_t] (f_h^t) = \mathbb{E}^{\mathbb{Q}} [X]$$

for all $f_h^t \in \mathcal{P}_t$.

Independent random variables III

In fact

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[X | \mathcal{P}_t](f_h^t) &= \sum_{m=1, \dots, M} x_m \cdot \mathbb{Q}[\{\omega : X(\omega) = x_m\} | f_h^t] \\&= \sum_{m=1, \dots, M} x_m \cdot \frac{\mathbb{Q}[\{\omega : X(\omega) = x_m\} \cap f_h^t]}{\mathbb{Q}[f_h^t]} \\&= \sum_{m=1, \dots, M} x_m \cdot \frac{\mathbb{Q}[\{\omega : X(\omega) = x_m\}] \cdot \mathbb{Q}[f_h^t]}{\mathbb{Q}[f_h^t]} \\&\quad \text{because } X \text{ is } \mathbb{Q}\text{-indep. of } \mathcal{P}_t \\&= \sum_{m=1, \dots, M} x_m \cdot \mathbb{Q}[\{\omega : X(\omega) = x_m\}] = \mathbb{E}^{\mathbb{Q}}[X]\end{aligned}$$

Martingales I

[DEF] The process $M = \{M(t)\}_{t=0}^T$ adapted to $\mathcal{P} = \{\mathcal{P}_t\}_{t=0}^T$ is a **martingale under the probability $\mathbb{Q}$** and with respect to the information structure $\mathcal{P}$ if, for any pair of dates $t_1 < t_2$ with $t_1, t_2 \in \{0, \dots, T\}$, we have

$$\mathbf{E}^{\mathbb{Q}}[M(t_2) | \mathcal{P}_{t_1}] = M(t_1) \quad (1)$$

Equivalently:

- M is a martingale if and only if

$$\mathbf{E}^{\mathbb{Q}}[M(t_2) - M(t_1) | \mathcal{P}_{t_1}] = 0 \text{ for any } t_1 < t_2$$

- M is a martingale if and only if

$$\mathbf{E}^{\mathbb{Q}}[M(t + \Delta t) - M(t) | \mathcal{P}_t] = 0 \text{ for any } t, \text{ and any } \Delta t$$

Martingales II

Moreover:

If $M = \{M(t)\}_{t=0}^T$ is a $\mathbb{Q}$ -martingale with respect to $\mathcal{P} = \{\mathcal{P}_t\}_{t=0}^T$ then, for all $t \in \{\Delta t, \dots, T\}$ we have that

$$\mathbf{E}^Q [M(t)] = M(0)$$